

Beyond Persistence? PAGE: Persistence-Anchored Gated Forecasting with Spatio-Temporal Graph Networks for Dengue

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Abstract—[WRITE: abstract, 180–220 words, written last]

Index Terms—dengue forecasting, spatio-temporal graph neural networks, persistence, SEIR, data leakage, prospective evaluation

I. INTRODUCTION

Dengue remains a major public-health problem in Sri Lanka, with recurrent outbreaks that place pressure on hospitals and vector-control services. The 2017 epidemic alone recorded more than 186,000 cases and 440 deaths [1]. Reliable district-level forecasts can support earlier preparation of clinical resources and targeted control activities. In this work, we consider weekly dengue forecasting for the 25 districts of Sri Lanka with a three-week prediction horizon.

Recent work on dengue forecasting follows two main directions. Spatio-temporal graph neural networks (ST-GNNs) represent districts as nodes and use their spatial relationships together with temporal case patterns; Weng et al. [2] evaluated several such architectures on Sri Lankan dengue data. A second direction introduces epidemic mechanisms into neural forecasting. For example, Liu et al. [3] use an LSTM to predict the force of infection of an SEIR model. These approaches suggest a natural combination in which a graph encoder captures spatial and temporal information while an SEIR model represents disease dynamics.

However, an important property of weekly dengue incidence is easily overlooked: the recent case count itself is a very strong predictor. On our corrected dataset, cases in week $t - 1$ explain $r^2 = 0.89$ of the variation in week t , whereas the best individual climate covariate at a causal lag explains only $r^2 = 0.03$. The simple persistence forecast,

$$\hat{y}_{t+h} = y_t, \quad (1)$$

is therefore a strong baseline. Existing ST-GNNs nevertheless predict the future case level directly from their learned representations. They must first learn the approximate identity relationship $\hat{y}_{t+1} \approx y_t$ before learning any useful correction to it. In our controlled evaluation, three of five published ST-GNNs fail to do this effectively and are 5–17 RMSE worse than persistence on validation. This weakness is difficult to observe in the existing benchmark because persistence is not reported as a baseline. We additionally find information leakage in the benchmark data, including data from 2023 placed in the training split while testing on 2018–2022, and a rainfall

feature aligned with future rather than past weeks. Existing mechanistic approaches also do not isolate the contribution of epidemic physics using an otherwise identical model without the simulator.

To address this problem, we propose PAGE (**P**ersistence-**A**nchored **G**ated **E**stimation), a lightweight forecasting head that can be attached to different spatio-temporal encoders. Instead of asking the neural network to reconstruct the complete future case level, PAGE begins with the persistence forecast and learns only a departure from it. A learned scalar gate, initialized close to zero ($g = 0.12$), controls how strongly this correction changes the persistence prediction. This gives the model a strong forecast at initialization while allowing it to move away from persistence when the data provide evidence to do so. The same framework also allows an SEIR simulator to generate the correction, enabling a matched experiment on whether epidemic physics provides additional predictive value.

Across the development experiments, PAGE reduces the validation RMSE of STGAT, A3TGCN, and DCRNN by 5.95–16.29 points, with the improvement observed in all nine paired runs for each encoder. With AAGCN, PAGE also beats persistence in all nine runs on both validation and test splits. Ablation experiments show that these gains come mainly from the persistence anchor and the learned gate rather than the SEIR simulator. This conclusion remains consistent in the later 2024–2026 evaluation: on 85 forecast origins, the SEIR and non-SEIR variants show no significant difference ($\Delta\text{RMSE} = -0.03$, $p = 0.67$). Persistence obtains the lowest RMSE on these later weeks (34.75 compared with 35.43 for PAGE), although PAGE achieves lower MAE and SMAPE.

The main contributions of this work are:

- 1) We propose PAGE, an encoder-independent persistence-anchored gated forecasting head that adds only one learned gating parameter. Controlled ablations separate the effects of the persistence anchor, the gate, and the SEIR component.
- 2) We audit the existing Sri Lankan dengue benchmark and construct a leakage-controlled weekly dataset from original source reports. The development dataset contains 559 weeks from 2013–2024 and is extended with later reports through August 2026 for prospective evaluation.
- 3) We provide a common evaluation framework for six neural encoders together with matched physics and non-physics controls, identical temporal splits, paired

TABLE I

COMPARATIVE ANALYSIS ON ONE HARNESS (CORRECTED DATA, 3 ORIGINS $\times$ 3 SEEDS). MEAN VALIDATION / TEST RMSE IN WEEKLY CASES. TOP: THE FIVE PUBLISHED ST-GNNs AND THE LSTM OF SEIR-LSTM, AS PUBLISHED (DIRECT HEAD), WITH PAGE, AND WITH PAGE PLUS THE SEIR BRANCH (THE SEIR-GNN, OR SEIR-LSTM FOR THE LSTM). SQUARED ERROR, NO SEASONAL FEATURES. BOTTOM: BASELINES; B IS THE CONFIGURATION VALIDATION SELECTS. SELECTION USES VALIDATION ONLY.

encoder	direct	PAGE	PAGE+SEIR
AAGCN	16.71 / 35.15	16.57 / 33.53	17.43 / 35.58
ASTGCN	16.83 / 35.61	16.76 / 34.76	17.51 / 35.71
LSTM	16.91 / 35.10	16.94 / 35.85	17.69 / 35.52
STGAT	22.93 / 62.00	16.98 / 36.46	17.63 / 35.84
A3TGCN	28.63 / 60.40	17.68 / 37.16	17.68 / 35.50
DCRNN	34.64 / 73.52	18.35 / 39.02	17.67 / 35.52
baseline	validation / test		
Persistence	17.86 / 36.02		
No message passing (res.)	16.90 / 34.57		
GCN (res.)	16.84 / 34.50		
STID-style MLP [4]	16.35 / 36.33		
k -NN analogues	16.52 / 34.34		
NB-GLM	16.97 / 42.85		
Boosted trees	17.94 / 38.96		
Boosted trees + climate	17.40 / 36.49		
B : AAGCN, NB, season	15.51 / 37.54		

experiments, and a later 2024–2026 evaluation. The results show that persistence is a stronger prior for this forecasting problem than the tested SEIR dynamics.

II. RELATED WORK

[WRITE: (a) ST-GNNs and the gap]

[WRITE: (b) mechanistic hybrids and the gap]

[WRITE: (c) persistence and simple baselines, link to PAGE]

III. DATASET

[WRITE: (a) what is wrong with the benchmark array]

[WRITE: (b) the corrected 559-week dataset]

[WRITE: (c) the 2024–2026 extension, with the amendment stated]

IV. THE PAGE FRAMEWORK

[WRITE: (a) problem formulation, symbols only]

[WRITE: (b) encoder]

[WRITE: (c) the PAGE head: equation and why it works]

[WRITE: (d) the optional SEIR branch, 2–3 sentences]

[WRITE: (e) loss]

[WRITE: (f) implementation details, with the numbers]

V. EXPERIMENTAL SETUP

[WRITE: (a) baselines]

[WRITE: (b) protocol, stated as a fair comparison]

[WRITE: (c) metrics and statistical tests]

VI. RESULTS

A. Comparative analysis

[WRITE: (a) Table encoders: who fails, how PAGE repairs, against persistence]

TABLE II

ABLATION: FROM THE PUBLISHED DIRECT HEAD TO PAGE. EACH ENCODER BEHIND FOUR HEADS THAT ADD, IN TURN, THE PERSISTENCE ANCHOR (RESIDUAL), THE LEARNED GATE (PAGE) AND THE SEIR BRANCH (MEAN VALIDATION RMSE; CONFIGURATION OF TABLE I). THE LAST TWO COLUMNS PAIR PAGE+SEIR WITH PAGE; NEGATIVE FAVOURS THE SEIR BRANCH, AND THE COUNT IS RUNS WHERE IT WINS.

encoder	direct	resid.	PAGE		SEIR branch	
			PAGE	+SEIR	val	test
AAGCN	16.71	16.72	16.57	17.43	+0.86 (1/9)	+2.05 (0/9)
ASTGCN	16.83	16.85	16.76	17.51	+0.75 (0/9)	+0.95 (3/9)
LSTM	16.91	16.96	16.94	17.69	+0.75 (0/9)	-0.33 (6/9)
STGAT	22.93	17.85	16.98	17.63	+0.66 (1/9)	-0.62 (8/9)
A3TGCN	28.63	17.47	17.68	17.68	-0.00 (6/9)	-1.66 (9/9)
DCRNN	34.64	17.52	18.35	17.67	-0.68 (3/9)	-3.50 (9/9)
persistence	17.86					

B. Ablation study

[WRITE: (b) Table rescue: anchor, gate, SEIR branch; why the pure SEIR decoder fails]

C. Hyperparameter tuning and computational cost

[WRITE: (c) tuning and (d) cost, a few sentences]

D. Prospective evaluation, 2024–2026

[WRITE: (e) Table prospective, stated honestly]

VII. DISCUSSION

[WRITE: why PAGE works, its limit, physics, limitations]

VIII. CONCLUSION

[WRITE: 3–4 sentences and the code link]

REFERENCES

- [1] H. A. Tissera, B. D. W. Jayamanne, R. Raut, S. M. D. Janaki, Y. Tozan, P. C. Samaraweera, P. Liyanage, A. Ghouse, C. Rodrigo, A. M. de Silva, and S. D. Fernando, “Severe dengue epidemic, sri lanka, 2017,” *Emerging Infectious Diseases*, vol. 26, no. 4, pp. 682–691, 2020.
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- [3] Y. Liu, P. Fransson, J. Heidecke, P. Liyanage, J. Wallin, and J. Rocklöv, “An explainable covariate compartmental model for predicting the spatio-temporal patterns of dengue in Sri Lanka,” *PLoS Computational Biology*, vol. 21, no. 9, p. e1013540, 2025.
- [4] Z. Shao, Z. Zhang, F. Wang, W. Wei, and Y. Xu, “Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting,” in *ACM International Conference on Information and Knowledge Management (CIKM)*, 2022.

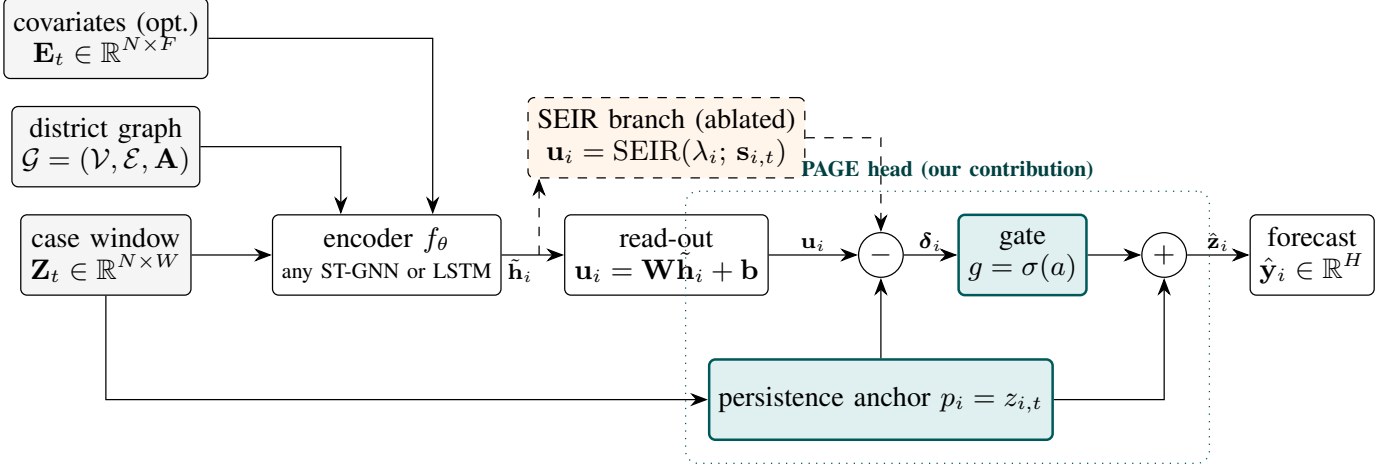


Fig. 1. PAGE. An unchanged encoder reads the last W weeks of standardised log cases on the district graph. Its read-out $\mathbf{u}_i$ is turned into a departure $\delta_i = \mathbf{u}_i - p_i \mathbf{1}$ from the persistence anchor, shrunk by a learned gate, and added back to the anchor. The dashed SEIR branch replaces the read-out with a differentiable simulator; it is the component the ablation removes.

TABLE III

PROSPECTIVE EVALUATION ON WEEKS NO MODEL OR SETTING HAD SEEN: 85 FORECAST STARTS, 2024-W11 TO 2026-W32. SETTINGS FROZEN ON DEVELOPMENT DATA BEFORE THE NEW WEEKS WERE PARSED. NEURAL ARMS: MEAN OVER 3 SEEDS, SEED RANGE OF RMSE IN BRACKETS. BOLD: BEST PER METRIC. PRE-REGISTERED TESTS (BLOCK BOOTSTRAP OVER FORECAST STARTS, HOLM OVER TWO): P1, SEIR BRANCH VS PAGE: $-0.03 [-0.41, +0.12]$, $p_{\text{HOLM}} = 0.67$; P2, PAGE+SEIR VS PERSISTENCE: $+0.65 [-0.99, +1.69]$, $p_{\text{HOLM}} = 0.62$.

arm	RMSE	MAE	SMAPE	MAPE	RMSE $h=1/2/3$
Persistence	34.75	11.78	50.3	53.3	15.4/35.7/46.0
PAGE (adaptive graph)	35.43 [35.24, 35.60]	11.35	45.7	43.9	16.0/35.7/47.3
PAGE + SEIR branch	35.40 [35.11, 35.56]	11.29	46.1	43.5	15.8/35.7/47.3
Residual head (no gate)	35.74 [35.37, 36.19]	11.61	46.6	43.0	16.4/36.0/47.6
AR(3) ridge	36.35	11.35	46.7	44.5	17.3/35.9/48.8
Seasonal naive	62.43	27.53	74.6	113.4	50.7/62.5/72.2